DEVELOPMENT_PLAN

Helios — Development Plan

Companion to: `DEPENDENCY_MAP.md` (the *what depends on what*) and Bible §12 (Claude Code workflows) + §23 (roadmap). · **Version:** v1.0 · **Date:** 2026-06-03

Purpose. This is the *how-we-build-it* plan: the milestones M0–M12 turned into concrete work packages, each with the Claude Code workflow to use, its definition-of-done, its exit gate, and what can run in parallel. It is written for **continuity and resume-ability** — a future session reads §2 (current status) + §12 (milestone tracker), finds the next ⬜ work package, opens its spec doc, and starts.

Status legend: ✅ done · 🔄 in progress · ⬜ todo. Update the tracker (§12) as packages land — it is the single resume point.

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1. How to use this plan

1. Open §12 **Milestone Tracker**; find the lowest-numbered milestone not ✅.
2. Open that milestone's **work-package table** (§7–§9) for the task list, the spec doc, the workflow, and the DoD.
3. Build in dependency-safe order (never start a package before its deps are green — the tracker enforces this).
4. Run the package's tests; meet its **exit gate**; mark it ✅; update `CLAUDE.md` §10 (current status).
5. Keep the three continuity docs in lockstep (CLAUDE.md, DEPENDENCY_MAP, this plan).

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2. Current status (2026-06-03)

Specification & continuity layer — DONE. The project is fully specified and the governance keystone is authored & validated:

Artifact	State
`docs/architecture/HELIOS_PROJECT_BIBLE.md` (25 sections)	✅
`docs/planning/DEPENDENCY_MAP.md`	✅
`CLAUDE.md`	✅
`models/semantic/semantic_layer.yaml` (A5.1, 47 metrics / 19 dims)	✅ authored + referential-integrity validated (0 dangling refs)
`docs/architecture/MCP_ARCHITECTURE.md` (A6.x spec)	✅
`docs/architecture/AGENT_ARCHITECTURE.md` (A8.x/A9.x spec)	✅

Code layer — not started. No repo scaffold, no GCP/dbt setup, no models, no servers, no agents, no eval. The semantic registry is *authored* but cannot be *compiled/served* until the marts it references exist (M3/M4).

Next action: M0 — repo scaffold + GCP/IAM/ADC + dbt config. (§7, WP-0.1.)

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3. Guiding execution principles

1. **Governed-first.** Build the data spine and the semantic layer before agents. No code path hand-authors SQL or computes stats (G1–G5).
2. **TDD on SQL.** Write the dbt/golden test *before* the model. Keystones (sessionization, funnel monotonicity, revenue reconciliation, `decompose_change`) get golden-value tests because they fail *silently*.
3. **Eval-driven dev (from M10).** Once the benchmark exists, no change merges that drops top-1 accuracy >2 pts or introduces any hallucination.
4. **Front-load keystones.** `int_ga4__sessionized` (A3.1), `int_ga4__funnel_steps` (A3.2), the registry (A5.1, done), and `stats-mcp.decompose_change` (A6.3) have the largest blast radius — build them carefully and early.

5. **Parallelize independent tracks** (§10): the data spine, the data-independent math (`stats` / `experiment`), and memory/finance can progress concurrently via subagents.
6. **Never start a milestone before its deps are green** (`dbt build` + tests; from M7, the loop runs; from M10, the eval gate).
7. **Docs in lockstep.** Adding/renaming an artifact updates DEPENDENCY_MAP + this tracker; a canonical-name change updates the Bible Reference Card + CLAUDE.md.

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4. The Claude Code workflow per activity (Bible §12, made concrete)

Activity	How to use Claude Code
Planning	Use **plan mode** to expand each milestone into a step plan *checked against* `DEPENDENCY_MAP.md` before editing. This file is the tracker.
Architecture	The specs exist (Bible, MCP_ARCHITECTURE, AGENT_ARCHITECTURE). For any *new* component, write/extend a spec doc first, then code to it.
Coding	Generate dbt models from the registry YAML; use **subagents** to build independent tracks in parallel (§10); reference the exact spec doc in each prompt.
Testing	TDD: write the dbt test / golden test first, then make it pass. Keystones get golden-value tests (§8).
MCP development	Scaffold each server from `MCP_ARCHITECTURE.md §6/§9` ; **dogfood** by registering the server and having Claude Code call its own tools; write contract tests (§11 of the MCP doc).
Documentation	Auto-generate `dbt docs` ; keep CLAUDE.md/DEPENDENCY_MAP/this plan current; draft the case-study writeup at L3 (§9).
Deployment	Claude Code writes the GitHub Actions CI (dbt build+test+eval) and the Cloud Scheduler entrypoint for autonomous runs (M10/M11).

4.1 Custom slash commands to create early (M0–M1)

Investing in these once pays off every package after:

- `/resume` — read the tracker + CLAUDE.md, report current state and the next package.
- `/new-metric <name>` — add a metric to `semantic_layer.yaml` + its test + bump version + re-run the registry validator.
- `/new-model <layer> <name>` — scaffold a dbt model + its `schema.yml` test stub.
- `/new-mcp-tool <server> <tool>` — scaffold a tool (signature, schema, error modes, contract test) per the MCP doc.
- `/new-scenario <id>` — add an eval scenario YAML + register it (M10).
- `/run-eval [--smoke|--full]` — run the harness, render the results table, compare to baseline.

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5. Indicative schedule (solo portfolio cadence; relative, not committed dates)

Effort sizes are relative (S ≈ ½–1 day, M ≈ 1–3 days, L ≈ 3–6 days, XL ≈ 1–2 weeks). Anchored to Bible §8 sizing (L1 ≈ 1–2 wks, L2 ≈ 3–5 wks, L3 ≈ 6–10 wks total).

Phase	Maturity	Milestones	Indicative	Headline outcome
P0	L1 (Intern MVP)	M0 → M7	~Sprint 1–2 (≈2 wks)	One anomaly → governed Decision Brief in <5 min, 0 hallucinated columns
P1	L2 (Strong portfolio)	M8 → M10	~Sprint 3–6 (≈4–6 wks)	7-agent loop + eval harness: **≥85% root-cause vs ≤45% baseline** in CI
P2	L3 (Top-1% undergrad)	M11	~Sprint 7–9 (≈3–4 wks)	Autonomous scheduled runs, forecasting/cohorts/RFM, full Critic battery, memory-driven learning
P3/P4	Production / Frontier	M12+	beyond	Multi-tenant, warehouse-agnostic, causal inference, auto-executed experiments

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6. Parallelization & critical path

Three tracks run concurrently after **M0** (DEPENDENCY_MAP §7); use subagents:

- **Track A — Data spine (critical path):** M1 → M2 → **M3 (keystone)** → M4 core → M5 (registry live) → M6 (semantic + warehouse MCP).
- **Track B — Math & experiment (data-independent, start day 1):** `stats-mcp` (incl. `decompose_change`), `experiment-mcp`, `report-mcp` core — each with unit/golden tests. Joins at M7/M9.
- **Track C — Memory & finance (independent):** `helios_memory` DDL + seeds, finance facts, `dim_date` / `dim_items`. Gates `report-mcp` memory tools (A7.4) and the full loop.

All three converge at **M7/M9 (agents)**. The schedule-determining chain is Track A → agents → eval; the **co-critical** memory branch (A7.1→A7.4→Critic/Narrator) must converge before the full eval run (DEPENDENCY_MAP §4).

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7. Phase 0 — L1 MVP (M0 → M7)

Goal: prove the governed spine end-to-end on one diagnosis path. **Exit (L1):** `reconcile('revenue','day')` matches a hand-written control query to the cent; 0 hallucinated columns; one anomaly → brief in <5 min.

WP	Milestone	Artifacts	Spec doc	Workflow	Definition of Done / tests	Size
WP-0.1	M0	A0.1, A0.3, A0.4, A0.5	Bible §16.7–16.8, §17	plan mode + scaffold	`dbt debug` connects; ADC authenticates; `mcp_servers.yaml` stub present	M
WP-0.2	M0	A0.2 CLAUDE.md	—	—	✅ **done**	—
WP-1.1	M1	A1.1–A1.6 macros/source/seed	Bible §16.3, §16.6	TDD	macros compile; `get_event_param` / `sessionize` / `channel_group_case` unit-tested; seed loads	M
WP-2.1	M2	A2.1–A2.3 staging	Bible §16.9, §8.1	TDD	`stg_ga4__events` / `stg_ga4__event_params` build as views; not_null/unique key tests pass	M
WP-3.1	M3 ★	A3.1 `int_ga4__sessionized`	Bible §8.5	TDD + golden	`session_key` unique; session-scoped source/medium derivation golden test (traffic_source fallback); engaged_session predicate	L
WP-3.2	M3 ★	A3.2 `int_ga4__funnel_steps`	Bible §10; CLAUDE.md §5	TDD + golden	`reached_*` max-downstream; **monotonicity golden test** (`sessions ≥ reached_view_item ≥ … ≥ reached_purchase`)	M
WP-4.1	M4	A4.1–A4.7 core marts	Bible §8.3–8.4	TDD	`fct_funnel` / `fct_daily_funnel` /dims build; **`revenue_reconciles` to the cent**; channel `accepted_values` (10 groups)	L
WP-4.2	M4 ‖	A4.8–A4.9 finance, A4.12 tests	Bible §8.3, §16.4	TDD	`fct_orders` / `fct_order_items` ; dedup by `transaction_id` ; net = gross − refund	M
WP-5.1	M5	A5.1 ✅ + A5.2 compile gate	Bible §14; SEMANTIC_LAYER.yaml	TDD	registry **compiles against the real marts**; referential-integrity check wired as a test; worked example (§14.7) golden SQL	S
WP-6.1	M6 ★	A6.0, A6.2 semantic-mcp, A6.1 warehouse-mcp	`MCP_ARCHITECTURE.md` §6.1–6.2, §9	MCP dev + dogfood	`build_query → dry_run → run_query → reconcile` round-trips; byte-budget gate fires; unknown-name → hard fail; contract tests	L
WP-6b.1	M6b ‖	A6.3 stats-mcp	`MCP_ARCHITECTURE.md` §6.3, §9; `AGENT_ARCHITECTURE.md` §6.4	TDD + golden	`decompose_change` **golden test** (mix=−0.0018, rate=0, int=0); significance vs scipy ref; determinism (seed)	L
WP-6b.2	M6b ‖	A6.4 experiment-mcp, A6.5 report-mcp core	`MCP_ARCHITECTURE.md` §6.4–6.5	TDD	`power_analysis` vs closed-form ref; `render_brief([])` → EmptyFindings; design rejects ungoverned metrics	M
WP-7.1	M7	A8.0 framework, A8.1 Monitor, A8.6 Narrator(min), A9.1 runner, A12.1 basic brief	`AGENT_ARCHITECTURE.md` §4, §6.2, §6.7, §13	plan mode + subagents	**L1 gate:** one anomaly → brief <5 min; 0 hallucinated columns; audit log proves every SQL came from semantic-mcp	L

8. Phase 1 — L2 Strong Portfolio (M8 → M10)

Goal: the full 7-agent plan-execute-critique loop + the decomposition core, statistically defended and **benchmarked**. **Exit (L2):** ≥85% root-cause accuracy vs ≤45% baseline; 100% of findings carry significance + dollar impact; cost under byte budget.

WP	Milestone	Artifacts	Spec doc	Workflow	Definition of Done / tests	Size
WP-8.1	M8	A7.1 memory DDL, A7.2 vector store, A7.3 seeds, A7.4 report memory tools	Bible §22; `MCP_ARCHITECTURE.md` §6.5	TDD	`save_diagnosis` → `recall_prior` round-trips (BigQuery + ANN); seasonality/launch calendars seeded for the dataset window	L
WP-9.1	M9	A8.2 Decompose, A8.3 Diagnose, A8.4 Critic, A8.5 Prescribe, A8.7 Orchestrator, A9.3 audit, A4.13 exposures	`AGENT_ARCHITECTURE.md` §5–§10	plan mode + subagents	FSM routes correctly (clean-run short-circuit; DOWNGRADE→re-query≤MAX; DROP→suppression); a real run yields PASS findings w/ significance + $ + experiment; hypothesis-tree pruning honored	XL
WP-10.1	M10 ★	A10.1–A10.7 eval harness, A11.1 CI	Bible §20; `AGENT_ARCHITECTURE.md` §12	eval-driven + deploy	injector + 50 labeled scenarios; scorers (rootcause/decomp/detection/dollars/hallucination/faithfulness); **CI gate green: top-1 ≥0.85, hallucination =0, cost ≤ budget**; results table in PR comment	XL

9. Phase 2 — L3 Top-1% Undergrad (M11) and beyond

Goal: autonomy, statistical depth, adversarial rigor that read as production thinking. **Exit (L3):** <5 min/run sustained on schedule; eval accuracy holds across all canonical dimensions; memory-driven learning demonstrable.

WP	Milestone	Artifacts	Spec	DoD / tests	Size
WP-11.1	M11	A9.2 scheduler; `forecast` / `cohort_retention` / `rfm_segment` wired into Diagnose; A4.11 cohorts; full Critic battery; A12.2 export, A12.3 drill-down	Bible §23.3, §22; AGENT §6.5	scheduled autonomous runs land briefs; forecast-residual anomaly detection; cohort/RFM segments feed the hypothesis tree; Critic catch-rate measured on injected adversarial cases	XL
WP-11.2	M11	Case study / writeup	Bible §24–§25	published case study: architecture + the 85%-vs-45% benchmark + 3 walked-through diagnoses	M
WP-12+	M12+	Productionization (multi-tenant, warehouse-agnostic adapters, streaming) → Frontier (causal inference, auto-executed experiments)	Bible §23.4–23.5	per-phase exit criteria (e.g. two warehouses pass identical reconcile tests; a causal estimate validated vs a held-out experiment)	—

10. Testing & CI strategy across phases

- **Pyramid:** schema tests (not_null/unique/accepted_values/relationships) → data tests (`revenue_reconciles` , funnel monotonicity, rate bounds) → unit/golden tests (sessionization, `decompose_change` , build_query SQL snapshot) → contract tests (MCP tools) → integration (the eval harness).

- **Keystone golden tests (write these first):** session-key uniqueness; source/medium derivation; funnel monotonicity; revenue-reconciles-to-the-cent; `decompose_change` identity + the §6.2 worked example; `build_query` §14.7 SQL snapshot; the hallucination AST check.
- **CI turns on at M5** (registry compile) and becomes the **regression firewall at M10** (eval gate). On every PR touching `models/`, `semantic/`, `agents/`, `eval/`: `dbt build` + tests + eval (12-scenario smoke on push, full 50 on PR to `main`). Gates: top-1 ≥0.85, decomp MAPE ≤0.10, hallucination =0 (hard), F1 ≥0.85, dollar error ≤0.15, faithfulness ≥0.95, regression ≤2 pts.

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11. Definition of Done per maturity level

- **L1 (M7):** governed spine runs end-to-end on one funnel; `reconcile` to the cent; 0 hallucinated columns; a single anomaly → a brief in <5 min.
- **L2 (M10):** seven agents + five MCP servers; Critic active; CI eval harness; significance + dollar impact on **every** finding; **≥85% vs ≤45%** on the labeled benchmark; cost under byte budget.
- **L3 (M11):** autonomous scheduled operation with memory-driven learning; forecasting/cohorts/RFM; full Critic refutation battery; accuracy holds across all canonical dimensions; published case study.

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12. Milestone tracker (the living resume point — update as packages land)

Milestone	Status	Key artifacts	Exit gate
Specs/docs	✅	Bible, DEPENDENCY_MAP, CLAUDE.md, semantic registry (A5.1), MCP_ARCHITECTURE, AGENT_ARCHITECTURE	all authored; registry validated (0 dangling refs)
M0 Foundation	⬜ ← **NEXT**	A0.1, A0.3, A0.4, A0.5	`dbt debug` connects; ADC auth
M1 Sources & macros	⬜	A1.1–A1.6	macros compile; seed loads
M2 Staging	⬜	A2.1–A2.3	staging tests pass
M3 Sessionization + funnel ★	⬜	A3.1, A3.2, A3.3	monotonicity test passes; session_key unique
M4 Marts	⬜	A4.1–A4.9, A4.12, A4.14	revenue reconciles to the cent; 10-channel `accepted_values`
M5 Registry live	🔄	A5.1 ✅ / A5.2 ⬜	registry compiles against real marts; CI compile gate
M6 Grounding MCP pair	⬜	A6.0, A6.2, A6.1	build_query→dry_run→run_query→reconcile round-trips
M6b Stats/exp/report MCP ‖	⬜	A6.3, A6.4, A6.5	`decompose_change` golden test passes
M7 Minimal loop (L1)	⬜	A8.0, A8.1, A8.6, A9.1, A12.1	**L1 DONE:** anomaly → brief <5 min; 0 hallucinations
M8 Memory + report tools	⬜	A7.1–A7.4	save/recall round-trips; seeds loaded
M9 Full agent loop	⬜	A8.2–A8.5, A8.7, A9.3, A4.13	PASS findings carry significance + $ + experiment
M10 Eval + CI (L2)	⬜	A10.1–A10.7, A11.1	**L2 DONE:** ≥85% vs ≤45%; hallucination 0; cost ≤ budget
M11 Autonomy & depth (L3)	⬜	A9.2, A4.11, A12.2, A12.3, case study	**L3 DONE:** <5 min/run on schedule; accuracy holds all dims
M12+ Prod / Frontier	⬜	per Bible §23.4–23.5	per-phase criteria

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13. Risk register

Risk	Likelihood	Impact	Mitigation
Sessionization / funnel flags silently wrong	Med	High	golden tests first (WP-3.1/3.2); monotonicity + reconcile gates before any agent consumes them
LLM agent produces plausible-but-wrong diagnoses	Med	High	grounding (G1–G5), deterministic stats, adversarial Critic, and the eval gate (≥85%) — the whole architecture is the mitigation
BigQuery cost overrun	Low	Med	mandatory `dry_run`; `maximum_bytes_billed` cap at 5 GiB; partition pruning; smoke vs full eval split
Registry ↔ marts drift	Med	High	CI referential-integrity compile (A5.2); registry filename pinned; one physical file (CLAUDE.md guardrail)
Scope creep into dashboard/chatbot	Med	Med	anti-product stance in CLAUDE.md; Decision Brief is the only surface; conversation is secondary and late (M11)
Dataset limits (3-mo window, cookie identity)	Certain	Low	explicitly deferred items (Bible §23.7): LTV/cross-device/causal to P4; document caveats; Critic flags cookie-based user metrics
Solo continuity loss between sessions	Med	Med	the three lockstep docs + `/resume` command + this tracker as the single resume point

14. Cadence (running it solo)

- **Per session:** `/resume` → pick the next ⬜ package → plan-mode the steps against DEPENDENCY_MAP → TDD it → meet the exit gate → mark ✅ → update CLAUDE.md §10.
- **Per milestone:** demo the headline outcome (even to yourself); keep a short changelog; ensure CI is green before advancing.
- **At L1/L2/L3:** record the demo + numbers for the case study and resume bullets (Bible §24–§25) while they're fresh.